q-adic Finite-Size Boundary Spectra

of Multiplicative Shifts of Finite Type

Exact General Remainders and Golden Cantor Boundaries

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Abstract

For a primitive finite zero–one adjacency matrix A and an integer radix $q \geq 2$, we determine the entire order-one finite-size boundary of the multiplicative shift constrained along the chains $i, qi, q^2i, \ldots$. Adding a site at cutoff N changes exactly one chain, so the logarithmic increment depends only on the valuation $\nu_q(N)$. Exact summation by parts then expresses the centered prefix count as a uniformly convergent real series in the residues $N \bmod q^v$. This series extends continuously to the inverse limit $\mathbb{Z}_q$, including composite q, and its image is exactly the complete accumulation set of the finite-size remainder.

For the binary golden adjacency, Binet's formula turns the boundary map into an alternating digit series. An exact inequality in $\mathbb{Q}(\sqrt{5})$ proves that every coefficient dominates its full tail. The image is therefore a strongly separated Cantor set, with Hausdorff and box dimensions both equal to $\log 2/(2\log\varphi)$. The same coefficient tails are the nonzero radial leading coefficients of the ordinary cutoff generating function at primitive dyadic roots; their density forces the unit circle to be a natural boundary. The multiplicative-shift object, chain product, leading entropy, leading dimensions, and valid prior boundary results remain prior-owned: the contribution is the exact subleading boundary and its golden geometric and analytic closure.

1 Introduction

Leading entropy records how fast a language grows, but it deliberately forgets the bounded arithmetic left by an integer cutoff. That loss is especially visible for multiplicative shifts. Their constraints do not run along consecutive sites; they run along the chains

$$i,\ qi,\ q^2i, \ldots, \qquad q \nmid i.$$

The standard chain product gives the leading term, yet a prefix ending at N cuts the chains at different depths according to the base-q digits of N. The natural question is therefore finite rather than asymptotic: after subtracting the entropy term, which order-one values can actually occur?

We answer this question exactly for every integer $q \geq 2$ and every finite primitive zero–one adjacency matrix A. Let $Z(N)$ be the number of admissible labelings of $\{1, \ldots, N\}$, let h be the leading prefix entropy, and set $E(N) = \log Z(N) - hN$. The inverse limit

$$\mathbb{Z}_q := \varprojlim_n \mathbb{Z}/q^n\mathbb{Z}$$

is used as a compact state space even when q is composite; the boundary map is real-valued, not q-adic-valued. We prove that $E(N)$ extends to a continuous function $E_{A,q} \colon \mathbb{Z}_q \to \mathbb{R}$ and that its image is precisely the complete set of subsequential limits of $E(N)$ as $N \to \infty$.

The binary golden adjacency exposes additional geometry. Its boundary function is a signed digit series whose coefficients alternate and dominate their entire future tails. This all-level estimate makes the real boundary image a strongly separated Cantor set and determines its Hausdorff and box dimensions. The same coefficient tails control an ordinary generating function at primitive dyadic roots, producing a dense family of nonzero radial leading coefficients and therefore a natural boundary.

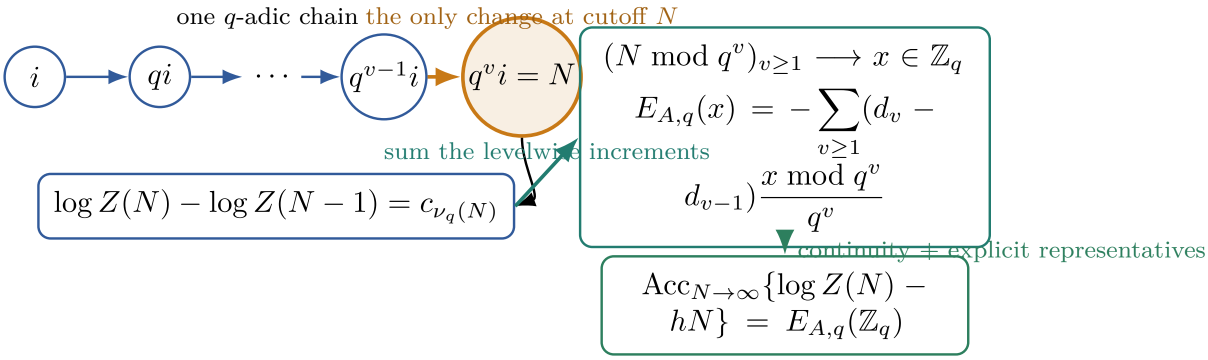

Figure 1: One cutoff step changes exactly one q-adic chain and hence exactly one valuation level. Summing these levelwise changes gives a uniformly convergent residue series; continuity and explicit diverging representatives identify its real image with the complete finite-size accumulation set.

Theorem contributions. The mathematical results are the following.

(T1) An exact finite-N increment and residue identity, followed by the equality

$$\mathrm{Acc}_{N\to\infty}\{\log Z(N) - hN\} = E_{A,q}(\mathbb{Z}_q).$$

(T2) For the binary golden control, a full-tail separation theorem and

$$\dim_H E_{A,2}(\mathbb{Z}_2) = \dim_B E_{A,2}(\mathbb{Z}_2) = \frac{\log 2}{2\log\varphi}.$$

(T3) For the ordinary cutoff series $G(z) = \sum_{N\geq 0} E(N)z^N$, the exact radial leading coefficient at every primitive dyadic root and the resulting unit-circle natural boundary.

Verification audit. Two physically distinct evaluators replay finite exact counts, residue identities, certified coefficient intervals, and cyclotomic controls. Their agreement tests implementations only. It bears no evidentiary weight for uniform convergence, the complete accumulation image, all-level separation, or the natural-boundary theorem; those statements are proved independently. The audit is reported retrospectively in section 6.

Ownership boundary. The multiplicative-shift object and its leading theory are established in earlier work [1, 4, 5]. Fibonacci word counts, the chain product, leading entropy and dimensions, and valid boundary-complexity results receive zero credit here. We also isolate a same-object displayed specialization in one author-manuscript version of Ban et al. [2]; the correction is a reconciliation, not a novelty claim about that source, and is stated with an explicit version-of-record caveat in section 2.

The paper proceeds from ownership and notation reconciliation (section 2) to the exact general boundary law (section 3), the golden Cantor theorem (section 4), and the natural boundary (section 5). The final sections separate finite verification from proof and record the nonclaims, Route outcome, and retrospective chronology.

2 Prior ownership and a same-object correction

Multiplicative shifts and leading geometry. The multiplicative golden-mean system and its leading counts and dimensions belong to the early literature [4]. The broader multiplicative-integer framework and leading Hausdorff/Minkowski theory were developed by Kenyon et al. [5]. Direct admissible-chain pattern products, chain-length densities, entropy, and leading error control are likewise prior-owned [1]. Recent affine extensions broaden that leading dimension theory [3]. We use these objects and formulas as inputs; none is presented as a contribution.

Table 1: Ownership subtraction. *Eligible* means only that the component remains in this theorem package after prior results are removed; it is not a priority assertion.

Component	Ownership/status	Candidate credit
Multiplicative SFT, q-adic chain partition, chain product	Primary-owned/elementary	zero
Fibonacci word counts; leading entropy and shift dimensions	Ban et al. [1], Fan et al. [4], Kenyon et al. [5]	zero
Boundary-complexity and surface-entropy framework	Ban et al. [2]	zero
Affine multiplicative-shift dimensions	Ban et al. [3]	zero
Exact valuation increment and continuous real boundary on $\mathbb{Z}_q$	proved here; no exact collision found in a bounded search	eligible, no priority claim
Golden full-tail separation, image dimension, radial coefficient	proved here for one frozen control	eligible, scoped

Digital fluctuations as a neighboring method family. Summatory functions of digit-additive sequences often carry periodic or fractal fluctuations. The literature represented by Madritsch [6] is therefore a methodological neighbor. Its real periodic fluctuations do not, by themselves, state the exact valuation increment, the continuous inverse-limit remainder, or the same-object accumulation theorem proved below. The distinction matters: our coordinate $x \bmod q^v$ is first selected in the finite quotient and then evaluated as a real number; no fractional-part substitution or q-adic-valued analytic function is involved.

Boundary complexity. Ban et al. [2] introduce boundary complexity and surface entropy for multiplicative integer systems. Their valid leading and boundary results, together with that terminology, remain prior-owned. The present paper asks a narrower question: an exact bounded remainder at every integer cutoff and the closure of all its subsequential values. The ownership separation is summarized in table 1.

2.1 The author-manuscript specialization

The checked artifact is the author manuscript arXiv:2210.09115v1, submitted 17 October 2022. Theorem 3.3(2) and Remark 3.4 in that artifact display, for the one-dimensional specialization at $N = p^{kn}$,

$$\log \left|\mathcal{P}([1, p^{kn}], X^p_{\Sigma_A})\right| - p^{kn} h = -(1 - p^{-1}) \log(\lambda_A)\, kn + o(kn). \tag{1}$$

The dictionary needed to compare (1) with our object is explicit:

author-manuscript symbol	present symbol	specialization	meaning
$X^p_{\Sigma_A}$	$X^{(q)}_A$	$p = q$	multiplicative SFT
$\left|\mathcal{P}([1, p^{kn}], X^p_{\Sigma_A})\right|$	$Z(N)$	$N = p^{kn}$	prefix count
λ_A	$\rho(A)$	same adjacency	Perron eigenvalue
h	h	same normalization	prefix entropy

Take the primitive full shift $A = J_d$, the $d \times d$ all-one matrix, with $d > 1$. Every site is free, hence

$$W_\ell = d^\ell, \qquad Z(N) = d^N, \qquad h = \log d.$$

The left side of (1) is therefore identically zero, whereas its displayed main term equals $-(1 - p^{-1}) \log d\, kn$, which is nonzero. More generally, the exact identity in theorem 3.1 gives $d_v = 0$ and $E(N) = 0$ for this control. Thus the displayed one-dimensional specialization cannot hold with the stated quantifiers. The bounded order-one identity below is the compatible same-object correction under our frozen primitive one-dimensional definition.

We checked only the author manuscript arXiv:2210.09115v1, submitted 17 October 2022. We did not line-check the version of record or any erratum, and make no claim about their displayed formulas.

Accordingly, we do not transfer (1) to the journal version. We preserve every valid leading and boundary result of Ban et al. [2], assign this correction zero prior-ownership credit, and make no first-discovery claim. A primary source containing the exact $\mathbb{Z}_q$ extension and complete accumulation theorem, or the same golden separation theorem, would trigger the live duplicate stop described in section 7.

3 Exact finite-N calculus and the q-adic boundary

Fix an integer $q \geq 2$ and a finite $d \times d$ primitive matrix $A \in \{0,1\}^{d\times d}$. On the alphabet $\mathcal{A} = \{1, \ldots, d\}$, define

$$X_A^{(q)} = \{x \in \mathcal{A}^{\mathbb{N}} : A_{x_n, x_{qn}} = 1 \text{ for every } n \geq 1\}.$$

The cutoff N always means the prefix $\{1, \ldots, N\}$; only constraints with both endpoints in this prefix are imposed. Write $Z(N)$ for its number of admissible labelings and set $Z(0) = 1$. For ordinary A-admissible words, put

$$W_0 = 1, \qquad W_\ell = \mathbf{1}^T A^{\ell-1}\mathbf{1} \quad (\ell \geq 1).$$

3.1 Chains and increments

Every positive integer has a unique representation $n = q^v i$ with $q \nmid i$. Consequently, the nonempty truncated chains

$$\mathcal{C}_i(N) = \{i, qi, \ldots, q^{\ell_i(N)-1} i\}, \qquad \ell_i(N) = 1 + \left\lfloor \log_q \frac{N}{i} \right\rfloor,$$

partition the prefix. Constraints never cross two such chains, so the prior-owned chain product is the exact identity

$$Z(N) = \prod_{\substack{1\leq i\leq N \\ q\nmid i}} W_{\ell_i(N)} = \prod_{\ell\geq 1} W_\ell^{C_\ell(N)}, \tag{2}$$

where the finite histogram is

$$C_\ell(N) = \left\lfloor \frac{N}{q^{\ell-1}} \right\rfloor - 2\left\lfloor \frac{N}{q^\ell} \right\rfloor + \left\lfloor \frac{N}{q^{\ell+1}} \right\rfloor.$$

Let $\nu_q(N) = \max\{v \geq 0 : q^v \mid N\}$. Adding the site N extends only the chain rooted at $N/q^{\nu_q(N)}$, from length $\nu_q(N)$ to length $\nu_q(N) + 1$. Define

$$c_v = \log \frac{W_{v+1}}{W_v}, \qquad \rho = \rho(A), \qquad d_v = c_v - \log \rho.$$

Then

$$\log Z(N) - \log Z(N-1) = c_{\nu_q(N)}. \tag{3}$$

The convention $W_0 = 1$ covers $\nu_q(N) = 0$, when a new one-site chain appears.

Primitivity supplies a Perron spectral gap. Thus for constants $C_0 > 0$, $0 < \theta < 1$, and an integer $m \geq 0$,

$$W_\ell = C_0\rho^{\ell-1}(1 + O(\ell^m\theta^\ell)).$$

It follows that $d_v = O(v^m\theta^v)$, and hence both $\sum_v |d_v|$ and $\sum_{v\geq 1} |d_v - d_{v-1}|$ converge. The natural mean increment is

$$h = \sum_{v\geq 0} \frac{q-1}{q^{v+1}} c_v = \log\rho + \sum_{v\geq 0} \frac{q-1}{q^{v+1}} d_v. \tag{4}$$

3.2 The exact residue identity

The number of $n \le N$ with valuation v is

$$A_v(N) = \left\lfloor \frac{N}{q^v} \right\rfloor - \left\lfloor \frac{N}{q^{v+1}} \right\rfloor.$$

For $r_v(N) = N \bmod q^v \in \{0, \dots, q^v - 1\}$, this becomes

$$A_v(N) = N\frac{q-1}{q^{v+1}} - \frac{r_v(N)}{q^v} + \frac{r_{v+1}(N)}{q^{v+1}}. \tag{5}$$

Summing (3) by valuations and substituting (5) gives the following result.

Theorem 3.1 (Exact q-adic finite-size boundary)**.** *For every $N \ge 1$,*

$$E(N) := \log Z(N) - hN = -\sum_{v\ge1}(d_v - d_{v-1})\frac{N \bmod q^v}{q^v}. \tag{6}$$

For $x \in \mathbb{Z}_q$, let $x \bmod q^v$ denote the canonical integer representative of its level-v coordinate. The series

$$E_{A,q}(x) := -\sum_{v\ge1}(d_v - d_{v-1})\frac{x \bmod q^v}{q^v} \tag{7}$$

converges uniformly to a continuous real-valued map on $\mathbb{Z}_q$. Moreover,

$$\operatorname{Acc}_{N\to\infty} E(N) = E_{A,q}(\mathbb{Z}_q), \tag{8}$$

$$\operatorname{Acc}_{N\to\infty}(Z(N)e^{-hN}) = \exp(E_{A,q}(\mathbb{Z}_q)). \tag{9}$$

Proof. Equation (3) yields $\log Z(N) = \sum_{v\ge0} c_v A_v(N)$. Substitute (5), separate $c_v = \log\rho + d_v$, and use $\sum_v A_v(N) = N$ together with (4). An index shift gives (6); it is justified by the absolute summability of (d_v).

For $x \in \mathbb{Z}_q$, each coordinate $x \bmod q^v$ is locally constant and $0 \le (x \bmod q^v)/q^v < 1$. The summable majorant $\sum_{v\ge1}|d_v - d_{v-1}|$ proves uniform convergence and continuity.

If $E(N_j) \to y$ with $N_j \to \infty$, compactness of $\mathbb{Z}_q$ provides a q-adically convergent subsequence $N_{j_k} \to x$. Continuity then gives $y = E_{A,q}(x)$. Conversely, for fixed $x \in \mathbb{Z}_q$, the explicit integers

$$N_j = (x \bmod q^j) + q^j$$

tend to infinity and converge to x in $\mathbb{Z}_q$. Hence $E(N_j) \to E_{A,q}(x)$, proving the reverse inclusion. Finally, exponentiation is continuous and injective on $\mathbb{R}$, which gives (9). □

Remark 3.2 (Type of the boundary)**.** The fractions $(x \bmod q^v)/q^v$ in (7) are evaluated in $\mathbb{R}$ after selecting compatible finite-quotient coordinates. Thus $\mathbb{Z}_q$ is a compact inverse-limit domain, not the codomain of the series. No field structure is used, so composite q is included.

Remark 3.3 (Full-shift and one-symbol controls)**.** For $A = J_d$, $W_\ell = d^\ell$, $c_v = \log d$, and $d_v = 0$. Therefore $Z(N) = d^N$ and $E_{A,q} \equiv 0$. The case $A = [1]$ gives the same zero boundary with $d = 1$. These exact controls detect a spurious chain multiplicity or an omitted W_0.

4 Golden boundary geometry

Specialize to

$$q = 2, \qquad A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

With $F_0 = 0, F_1 = 1$, the word counts are $W_\ell = F_{\ell+2}$. Set

$$t = \varphi^{-2} = \frac{3-\sqrt{5}}{2}, \qquad r = -t.$$

Binet's formula in the form

$$F_n = \frac{\varphi^n}{\sqrt{5}}(1 - r^n)$$

gives

$$d_v = \log \frac{1 - r^{v+3}}{1 - r^{v+2}}. \tag{10}$$

Every $x \in \mathbb{Z}_2$ has a unique expansion $x = \sum_{k\geq 0} \varepsilon_k 2^k$, with $\varepsilon_k \in \{0,1\}$. Substituting $x \bmod 2^v = \sum_{k=0}^{v-1} \varepsilon_k 2^k$ into (7) and reordering an absolutely convergent double series yields

$$E_{A,2}(x) = \sum_{k\geq 0} \gamma_k \varepsilon_k, \qquad \gamma_k = -\sum_{v\geq k+1} (d_v - d_{v-1}) 2^{k-v}. \tag{11}$$

Expanding the logarithms in (10) gives a second exact formula:

$$\gamma_k = \sum_{m\geq 1} a_m r^{m(k+2)}, \qquad a_m = \frac{(1-r^m)^2}{m(2-r^m)} > 0. \tag{12}$$

The second expression separates one dominant signed mode from a certified tail.

Theorem 4.1 (Golden strong separation and dimension)**.** *The coefficients in* (11) *satisfy*

$$\operatorname{sgn}(\gamma_k) = (-1)^k, \qquad |\gamma_k| > \sum_{j>k} |\gamma_j|, \qquad \frac{\gamma_{k+1}}{\gamma_k} \longrightarrow -\varphi^{-2}.$$

Consequently

$$K := E_{A,2}(\mathbb{Z}_2) = \left\{ \sum_{k\geq 0} \varepsilon_k \gamma_k : \varepsilon_k \in \{0,1\} \right\}$$

is a Cantor set, and

$$\dim_H K = \dim_B K = \frac{\log 2}{2 \log \varphi}.$$

Proof. The algebraic number t satisfies $0 < t < 1/2$ and $t^2 - 3t + 1 = 0$. For $m \geq 2$,

$$a_m \leq \frac{K}{m}, \qquad K = \frac{(1+t^2)^2}{2-t^2} = -6 + 3\sqrt{5}.$$

Define

$$S = \sum_{m\geq 2} \frac{a_m t^{2m}}{1-t^m}.$$

The elementary bounds $1/(1-t^m) \leq 1/(1-t^2)$ and $1/m \leq 1/2$ give the exact certificate

$$\begin{aligned} S &\leq \frac{Kt^4}{2(1-t^2)^2} = \frac{-87+39\sqrt{5}}{20} \\ &< a_1 t^3 = \frac{280 - 125\sqrt{5}}{11}. \end{aligned} \tag{13}$$

The strict comparison is certified without decimals:

$$a_1 t^3 - \frac{Kt^4}{2(1-t^2)^2} = \frac{6557 - 2929\sqrt{5}}{220} > 0, \qquad 6557^2 - 5 \cdot 2929^2 = 99044 > 0.$$

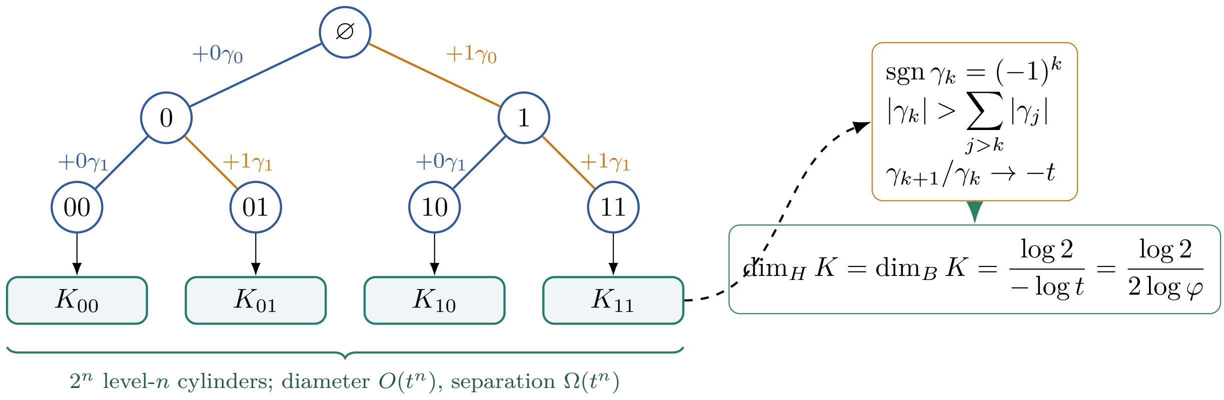

Figure 2: The golden digit map. The exact all-level inequality $|\gamma_k| > \sum_{j>k} |\gamma_j|$ separates siblings at every depth. With 2^n cylinders of diameter and separation comparable to t^n, the boundary image has dimension $\log 2/(-\log t)$.

Write $\gamma_k = a_1 r^{k+2} + R_k$, where the modes $m \geq 2$ form R_k. Equation (13) implies, for every $k \geq 0$,

$$|R_k| + \sum_{j>k} |R_j| \leq t^k S < a_1 t^{k+3}.$$

The first mode alone has the exact gap

$$a_1 t^{k+2} - \sum_{j>k} a_1 t^{j+2} = a_1 t^{k+2} \frac{1-2t}{1-t} = a_1 t^{k+3}.$$

Subtracting the higher-mode error proves full-tail domination. It also fixes the sign, while geometric decay of every $m \geq 2$ mode proves the ratio limit.

If two binary digit sequences first differ at k, their images differ by at least $|\gamma_k| - \sum_{j>k} |\gamma_j| > 0$. The digit map is therefore a continuous injection from compact $\mathbb{Z}_2$ into $\mathbb{R}$, hence a homeomorphism onto K. This proves the Cantor topology.

The ratio limit and the uniform gap give constants $C_1, C_2 > 0$ such that level-n cylinders have diameter at most $C_2 t^n$ and distinct level-n cylinders are separated by at least $C_1 t^{n-1}$. There are 2^n such cylinders. They give the Hausdorff and upper-box upper bound $s_0 = \log 2/(-\log t)$. For the reverse bound, push the fair Bernoulli measure to K. Every level-n cylinder has mass 2^{-n}, and an interval of radius comparable to t^n meets only a bounded number of them. Thus $\mu(I) \leq C|I|^{s_0}$. The mass-distribution principle and the same separation estimate give the Hausdorff and lower-box lower bounds. Since $t = \varphi^{-2}$, $s_0 = \log 2/(2\log\varphi)$. □

Remark 4.2 (Which dimension is computed)**.** The set $K \subset \mathbb{R}$ is the image of the finite-size boundary map. It is not the original multiplicative shift, and its dimension is not a restatement of the leading shift dimensions in [4, 5]. Nor does the argument analyze ordinary Minkowski content over all continuous covering scales.

5 Dense radial singularities

Keep the binary golden control and set $E(0) = 0$. The variable in

$$G(z) = \sum_{N \geq 0} E(N) z^N$$

marks the prefix cutoff N. Because theorem 3.1 makes $E(N)$ bounded, G is analytic on the open unit disk.

For an integer $Q \geq 2$, periodicity gives the rational residue series

$$R_Q(z) := \sum_{N\geq 0} (N \bmod Q) z^N = \frac{P_Q(z)}{1 - z^Q}, \qquad P_Q(z) = \sum_{a=0}^{Q-1} a z^a. \tag{14}$$

Writing $\Delta_v = d_v - d_{v-1}$, absolute convergence of $\sum_v |\Delta_v|$ permits the exact rearrangement

$$G(z) = -\sum_{v\geq 1} \frac{\Delta_v}{2^v} R_{2^v}(z), \qquad |z| < 1. \tag{15}$$

Theorem 5.1 (Radial coefficients and natural boundary)**.** *Let ξ be a primitive 2^v-th root of unity, $v \geq 1$. Then*

$$\lim_{s\uparrow 1} (1 - s) G(s\xi) = -\frac{\gamma_{v-1}}{2^{v-1}(1 - \xi)} \neq 0. \tag{16}$$

The unit circle is a natural boundary for G.

Proof. If $w < v$, then $1 - \xi^{2^w} \neq 0$, so the radial leading coefficient of R_{2^w} vanishes. If $w \geq v$, set $Q = 2^w$. Since $\xi^Q = 1$ and $\xi \neq 1$, differentiating the finite geometric sum gives

$$P_Q(\xi) = -\frac{Q}{1 - \xi}.$$

Together with $1 - s^Q \sim Q(1 - s)$, this implies

$$\lim_{s\uparrow 1} (1 - s) R_Q(s\xi) = -\frac{1}{1 - \xi}. \tag{17}$$

The limit may pass through (15). Indeed,

$$\frac{1 - s}{Q} |R_Q(s\xi)| \leq \frac{1 - s}{Q} \sum_{N\geq 0} Q s^N = 1,$$

so the level-w contribution is dominated by $|\Delta_w|$. Dominated convergence now yields

$$\lim_{s\uparrow 1} (1 - s) G(s\xi) = \frac{1}{1 - \xi} \sum_{w\geq v} \frac{\Delta_w}{2^w}.$$

By (11), $\sum_{w\geq v} \Delta_w / 2^w = -\gamma_{v-1}/2^{v-1}$, which proves (16); nonvanishing follows from theorem 4.1.

Primitive dyadic roots are dense on the unit circle. If G admitted analytic continuation across any boundary point, the continuation domain would contain an open boundary arc and hence a primitive dyadic root ξ. The identity theorem would make that continuation agree with G on the portion inside the disk and therefore remain finite as $s\xi \to \xi$. Equation (16) instead makes $G(s\xi)$ unbounded like a nonzero multiple of $(1 - s)^{-1}$, a contradiction. □

Normalization control. The first nonreal control prevents a silent phase error. For $Q = 4$ and $\xi = i$,

$$P_4(i) = -2 - 2i = -\frac{4}{1 - i}, \qquad \lim_{s\uparrow 1} (1 - s) R_4(si) = -\frac{1}{1 - i} = \frac{-1 - i}{2}.$$

This differs from $i/(1 - i) = (-1 + i)/2$. The exact coefficient in (16), rather than a merely proportional expression, is therefore essential.

Remark 5.2 (Radial is not meromorphic)**.** Equation (16) is an Abelian radial statement. Because the singular boundary points are dense, they are not isolated poles of a meromorphic continuation. The conclusion is precisely the natural boundary and no stronger meromorphic assertion.

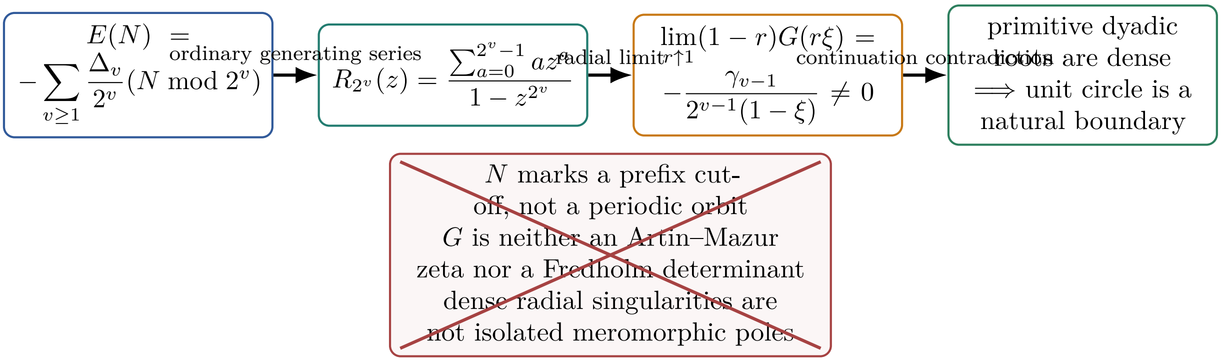

Figure 3: The analytic implication and its type firewall. Residue generating functions yield nonzero radial leading coefficients at a dense set of dyadic roots, which obstruct continuation across every boundary arc. The marker is a prefix cutoff: no orbit zeta, trace, determinant, or isolated-pole interpretation is used.

6 Exact computational replay

The computational layer checks finite consequences of the theorem package without supplying any step of its infinite proofs. Evaluator A enumerates labelings on literal source-graph components. Evaluator B independently expands the neutral inputs and uses the closed chain histogram, exact word counts, a positive Binet interval series, and cyclotomic quotient arithmetic. Their sources have distinct hashes, share no project-local imports, fixtures, or expected tables, and are compared by strict recursive type and value equality.

The finite cases include the exact golden prefix ledger

$$Z(0), \ldots, Z(10) = 1, 2, 3, 6, 10, 20, 30, 60, 96, 192, 288$$

and the one-symbol and full-shift zero boundaries. Coefficient intervals are certified with analytic tails rather than fitted from target values. The proof auditor, not either evaluator, owns the four infinite obligations listed in table 2.

The literal evidence boundary stored by the comparison is

$$\text{FINITE\_RESULTS\_DO\_NOT\_PROVE\_INFINITE\_THEOREMS.}$$

Accordingly, the audit supports reproducibility and catches source, normalization, residue, interval, analytic-type, and ownership mutations; it does not upgrade numerical agreement into a proof or a novelty claim.

7 Scope, chronology, and conclusion

Scientific limits. The primitive hypothesis is essential to the supplied Perron-gap proof. We do not infer the same theorem for reducible, irreducible-periodic, countable-state, or higher-step presentations. The Cantor and natural-boundary theorems belong only to the binary golden control, not to all q, A. The dimension is that of the real boundary image $E_{A,2}(\mathbb{Z}_2)$, not the original multiplicative shift, and no ordinary continuous-scale Minkowski-content statement is made. Finally, G is an ordinary prefix-cutoff generating function: it is neither an Artin–Mazur zeta function nor a determinant or transfer trace.

Route outcome. The exact strict Route tuple is

$$[\text{A0\_FAIL}, \text{A1\_FAIL}, \text{A2\_FAIL}, \text{A3\_PARTIAL\_ANALYTIC\_STRUCTURE}, \text{A4\_FAIL}],$$

with overall verdict ROUTE_A_REJECTED. Route B is locked. The single positive coordinate, A3_PARTIAL_ANALYTIC_STRUCTURE, records only the golden natural boundary. It does

Table 2: Canonical post-output block, transcribed from the protected authority artifacts identified in appendix B.3. Every value is retrospective. The block is an implementation audit only; it is not evidence for theorems 3.1, 4.1 and 5.1.

Field	Exact stored value
Finite comparison	580 cases = 548 theorem-domain +32 scope rejections; strict recursive type/value equality = true; projection SHA-256 `a4d487bafa038c65526514f4f99811cb765dc390b7f952bdb651b963838ecaca`.
Golden intervals and algebra	33 independently certified interval pairs, all overlapping; exact square-difference control $99044 > 0$.
Positive controls	one-symbol `PASS`; full shifts `PASS`; golden prefix ledger `PASS_10_ROWS`.
Evaluator source/output hashes	source A `87a71880cf49ea66fd4a9b71bfeef8466b6db60fe7597926c3767ff456f4d0a6`; source B `3c57736972f857b8fc239b58d9fe95b6900650dae1f6ff761662744b9f1d4fc5`; output A `0871d9c05f8bb32e05544ae2a690be3a2b0f052398f2efed3cd4fe180d1f1e6b`; output B `0b6f41f86008e48679a70e1485cb18a36463b8735baf440cfc6054d09b8898c9`.
Adversarial replay	19 frozen mutation families, 20 concrete instances, 52 designated consumer invocations, 0 survivors; 8 frozen external-auditor mutations, 0 survivors.
Infinite-proof audit	uniform Perron majorant, reverse accumulation inclusion, all-level separation certificate, and dominated Abelian passage all certified; finite-grid-as-proof = false.
Source boundary	Ban–Hu–Lai author-manuscript correction excerpt SHA-256 `b7c4aaf6c75e5a1790fc17f311242a8c56d6d23fd153657baed7dd93421c022f`; version-of-record line-checked = false; leading-result novelty credit = 0.
Route	primary 6/6, independent 6/6; tuple [A0_FAIL, A1_FAIL, A2_FAIL, A3_PARTIAL_ANALYTIC_STRUCTURE, A4_FAIL]; verdict ROUTE_A_REJECTED; Route B invocation = false; external STOP_DUPLICATE remains live conditional and is not a Route terminal.
Ledger and integrity	result ledger 22 entries, SHA-256 `2397a01fb66e42980b9a71a6861350afe52ceb258d3ae119dcbeef58b4d3086e`; final integrity 15/15; science SHA-256 `c988d083cbdd05a84f698497bfb6b8c13f9540fe3801497da9c7ce05d09f8e10`.
Authority state	State A; source commit `PREAUTHORITY_NO_COMMIT`; code commit `NONE`; authority writes 0; Git operations 0; paper manifest absent.

not create a periodic-orbit ledger, completed divisor, determinant comparison, or fixed self-adjoint Hilbert–Pólya operator. The external STOP_DUPLICATE condition remains live but is not a Route terminal: locating a primary source with the exact theorem and quantifiers would end a standalone publication claim without invalidating the internal derivation.

Retrospective chronology. The present candidate was chosen after the multiplicative-SFT literature, the exact remainder identity, the golden Binet control, the source subtractions, and the failures of earlier candidate assignments were known. Its survival after subtraction is not prospective selection, outcome-independent ranking, novelty, priority, or authorization. A proportional expression in an earlier Phase-2 note also used $\xi/(1-\xi)$ for the residue bracket. Direct evaluation gives $-1/(1-\xi)$, as shown in section 5; the corrected phase changes the coefficient but not nonvanishing or the natural-boundary conclusion. The paper uses only the corrected exact normalization.

Conclusion. Once prior ownership is removed, the remaining structure is a precise finite-size theorem. A one-site update reads the valuation of the cutoff; the summable deviations of ordinary word-count ratios turn those updates into a continuous real function on $\mathbb{Z}_q$; and explicit diverging representatives show that no boundary value is missed. In the golden case, the same coefficients control both a strongly separated Cantor image of dimension $\log 2/(2\log\varphi)$ and dense radial obstruction to analytic continuation. These conclusions are exact, scoped, and independent of the retrospective finite replay that checks their implementations.

A Proof details

This appendix records the algebra suppressed from the main-text proofs. Every infinite conclusion is derived here or in theorems 3.1, 4.1 and 5.1; finite computations are not used to close a limiting argument.

A.1 Chain census and Perron decay

Lemma A.1 (Exact chain histogram)**.** *For every $N \geq 1$ and $\ell \geq 1$, the number of roots i with $q \nmid i$ whose truncated chain has length exactly ℓ is*

$$C_\ell(N) = \left\lfloor \frac{N}{q^{\ell-1}} \right\rfloor - 2\left\lfloor \frac{N}{q^{\ell}} \right\rfloor + \left\lfloor \frac{N}{q^{\ell+1}} \right\rfloor .$$

Moreover, $\sum_{\ell\geq 1} \ell C_\ell(N) = N$.

Proof. The roots with chain length at least ℓ are the $i \leq N/q^{\ell-1}$ not divisible by q. Their number is

$$\left\lfloor \frac{N}{q^{\ell-1}} \right\rfloor - \left\lfloor \frac{N}{q^{\ell}} \right\rfloor .$$

Subtracting the corresponding count at level $\ell+1$ gives the formula. Every site belongs to one chain, so summing chain lengths counts $\{1, \dots, N\}$ once. □

Lemma A.2 (Summable Perron deviations)**.** *For primitive A, the sequence $d_v = c_v - \log\rho(A)$ satisfies*

$$\sum_{v\geq 0} |d_v| < \infty, \qquad \sum_{v\geq 1} |d_v - d_{v-1}| < \infty.$$

Proof. Let u, w be positive right and left Perron vectors, normalized by $w^T u = 1$. Finite-dimensional Jordan decomposition gives constants $C_1 > 0$, $0 < \theta < 1$, and $m \geq 0$ for which

$$A^{\ell-1} = \rho^{\ell-1} u w^T + O\Big(\ell^m (\theta\rho)^\ell\Big)$$

in any fixed matrix norm. Multiplication by $\mathbf{1}^T$ and $\mathbf{1}$ yields

$$W_\ell = (\mathbf{1}^T u)(w^T \mathbf{1})\rho^{\ell-1}\Big(1 + O(\ell^m\theta^\ell)\Big),$$

where the leading coefficient is positive. For sufficiently large ℓ, the relative error has modulus below $1/2$; the logarithm is Lipschitz on that interval. Hence

$$\log\frac{W_{v+1}}{W_v} = \log\rho + O(v^m\theta^v).$$

The first series follows after absorbing finitely many initial terms, and the second follows from $|d_v - d_{v-1}| \leq |d_v| + |d_{v-1}|$. □

A.2 Exact summation by parts

Starting from (3), group cutoffs according to their valuation:

$$\log Z(N) = \sum_{v\geq 0} c_v A_v(N).$$

Only finitely many $A_v(N)$ are nonzero. Equation (5) and $\sum_v A_v(N) = N$ give

$$\log Z(N) - hN = \sum_{v\geq 0} d_v \left(-\frac{r_v(N)}{q^v} + \frac{r_{v+1}(N)}{q^{v+1}} \right),$$

with $r_0(N) = 0$. By lemma A.2, the index shift is absolutely convergent, so

$$\log Z(N) - hN = -\sum_{v\geq 1} (d_v - d_{v-1}) \frac{r_v(N)}{q^v}.$$

This also gives a coefficientwise check. If $p_j = (q-1)/q^{j+1}$, then the coefficient of the formal variable d_j on the left is $A_j(N) - Np_j$. On the right it is

$$\begin{cases} r_1(N)/q, & j = 0, \\ -r_j(N)/q^j + r_{j+1}(N)/q^{j+1}, & j \geq 1, \end{cases}$$

and (5) makes the two expressions identical for every j.

A.3 Both accumulation inclusions

The inverse limit $\mathbb{Z}_q$ is a closed subset of the product of the finite discrete spaces $\mathbb{Z}/q^v\mathbb{Z}$; it is compact. The partial sums in (7) are continuous because each coordinate is locally constant. The majorant from lemma A.2 is independent of x, so the Weierstrass test gives a continuous uniform limit.

Suppose $E(N_j) \to y$ along $N_j \to \infty$. Compactness supplies a further subsequence $N_{j_k} \to x$ in $\mathbb{Z}_q$. Since $E(N) = E_{A,q}(N)$ for positive integer states, continuity gives $y = E_{A,q}(x)$. This proves one inclusion without assuming that the original integer sequence converges q-adically.

For the other inclusion, fix $x \in \mathbb{Z}_q$ and define

$$N_j = (x \bmod q^j) + q^j.$$

Then $N_j \geq q^j \to \infty$. For fixed v and $j \geq v$,

$$N_j \equiv x \bmod q^j \equiv x \bmod q^v \pmod{q^v},$$

so $N_j \to x$ in the inverse limit. Continuity proves $E(N_j) \to E_{A,q}(x)$. These two arguments establish (8); applying the real exponential establishes (9).

A.4 Derivation of the golden modes

Let $\Delta_v = d_v - d_{v-1}$. Substitution of the binary digits into (7) is justified by

$$\sum_{v\geq 1} |\Delta_v| 2^{-v} \sum_{k=0}^{v-1} 2^k \leq \sum_{v\geq 1} |\Delta_v| < \infty.$$

It yields (11). Define

$$B_k = \sum_{j\geq 1} \frac{d_{k+j}}{2^j}.$$

An index shift in (11) gives

$$\gamma_k = \frac{1}{2}(d_k - B_k). \tag{18}$$

Since $|r| < 1$,

$$d_k = \log(1 - r^{k+3}) - \log(1 - r^{k+2}) = \sum_{m\geq 1} \frac{1 - r^m}{m} r^{m(k+2)}.$$

For each m,

$$\sum_{j\geq 1} \frac{r^{mj}}{2^j} = \frac{r^m}{2 - r^m}.$$

Absolute convergence permits the j, m sums to be interchanged in B_k. Substituting the last two identities into (18) gives

$$\gamma_k = \sum_{m\geq 1} \frac{(1 - r^m)^2}{m(2 - r^m)} r^{m(k+2)},$$

which is (12).

A.5 All-level separation and dimension

For completeness, we verify every algebraic comparison in (13). Since $|1 - r^m| \leq 1 + t^m \leq 1 + t^2$ and $2 - r^m \geq 2 - t^2$ for $m \geq 2$,

$$a_m \leq \frac{1}{m} \frac{(1 + t^2)^2}{2 - t^2} = \frac{K}{m}.$$

Therefore
$$S \le \frac{K}{1-t^2}\sum_{m\ge 2}\frac{t^{2m}}{m} \le \frac{K}{1-t^2}\frac{1}{2}\sum_{m\ge 2} t^{2m} = \frac{Kt^4}{2(1-t^2)^2}.$$
Reducing in $\mathbb{Q}(\sqrt{5})$ gives the two displayed values in (13). Both 6557 and $2929\sqrt{5}$ are positive, and
$$6557^2 - (2929\sqrt{5})^2 = 99044 > 0,$$
so the difference is strictly positive.

For $R_k = \sum_{m\ge 2} a_m r^{m(k+2)}$,
$$|R_k| + \sum_{j>k} |R_j| \le \sum_{m\ge 2} \frac{a_m t^{m(k+2)}}{1-t^m} \le t^k S < a_1 t^{k+3}.$$
The first-mode gap is also $a_1 t^{k+3}$, because $1-2t = t(1-t)$. The difference is positive at every k, proving strong separation, while $|R_k| < a_1 t^{k+2}$ proves $\operatorname{sgn}\gamma_k = (-1)^k$. Dividing by $a_1 r^{k+2}$ and applying dominated convergence to the modes $m \ge 2$ proves $\gamma_{k+1}/\gamma_k \to r$.

The ratio limit lets us choose $b_1, b_2 > 0$ with
$$b_1 t^k \le |\gamma_k| \le b_2 t^k$$
for every k, after modifying the constants for finitely many initial indices. The level-n tail diameter is at most $C_2 t^n$, while the first-difference argument gives separation at least $C_1 t^{n-1}$. For $s > s_0 = \log 2/(-\log t)$, the 2^n cylinders have total s-content at most
$$2^n (C_2 t^n)^s = C_2^s (2t^s)^n \longrightarrow 0.$$
This proves the Hausdorff upper bound and the upper box bound.

Push the fair Bernoulli measure on $\{0,1\}^{\mathbb{N}}$ to K. Choose n so that $t^{n+1} < R \le t^n$. Separation implies that an interval of radius R meets at most a constant number M of level-n cylinders. Hence
$$\mu(I) \le M 2^{-n} = M(t^n)^{s_0} \le M t^{-s_0} R^{s_0}.$$
The mass-distribution principle gives $\dim_H K \ge s_0$, while the same separation gives the lower box bound. This completes the proof of theorem 4.1.

A.6 Interchange and dominated radial passage

For $|z| < 1$,
$$\sum_{v\ge 1} \frac{|\Delta_v|}{2^v} \sum_{N\ge 0} (N \bmod 2^v)|z|^N \le \frac{1}{1-|z|} \sum_{v\ge 1} |\Delta_v| < \infty.$$
Thus the N, v sums may be interchanged to obtain (15). At a primitive 2^v-th root ξ, the finite geometric derivative gives
$$P_Q(\xi) = \sum_{a=0}^{Q-1} a\xi^a = -\frac{Q}{1-\xi} \qquad (Q = 2^w,\ w \ge v).$$
For $s \in (0,1)$,
$$(1-s)\frac{|R_Q(s\xi)|}{Q} \le 1.$$
After multiplication by $|\Delta_w|/2^w = |\Delta_w|/Q$, this supplies the summable dominating sequence $(|\Delta_w|)$. Dominated convergence therefore proves (16). Its coefficient is nonzero by strong separation, and density plus the identity theorem gives the continuation contradiction in theorem 5.1.

Table 3: The objects used by the theorem have distinct types.

Symbol	Type and meaning	Forbidden identification
$X_A^{(q)}$	one-sided multiplicative SFT	additive shift or orbit ledger
N	positive prefix cutoff	orbit period or return time
$Z(N)$	finite admissible prefix count	fixed-point count
$x \in \mathbb{Z}_q$	inverse-limit residue state	real fractional part or a field element for composite q
$E_{A,q}(x)$	real order-one boundary value	point of the original shift
z in $G(z)$	ordinary cutoff marker	zeta, norm, roof, or trace marker
ξ	dyadic root used in a radial limit	isolated meromorphic pole

B Types, provenance, and reproducibility

B.1 Object and marker firewall

The prefix convention is fixed as $\{1, \ldots, N\}$, and the source edge is $n \mapsto qn$. Replacing the prefix by $\{0, \ldots, N-1\}$, replacing the edge by $n \mapsto n+q$, or imposing an edge whose second endpoint lies outside the prefix changes (3). Likewise, Acc means subsequential limits along cutoffs tending to infinity, not the closure of values below a fixed bound.

B.2 Source reconciliation and bounded search

The source audit assigned zero novelty credit to the object, chain decomposition, product, Fibonacci counts, leading entropy and dimensions, boundary-complexity terminology, and valid results of Ban et al. [1, 2, 3], Fan et al. [4], Kenyon et al. [5]. The digital-summatory literature was searched as a method neighbor [6]. The bounded search did not locate the exact theorem package, but negative search evidence was graded at most B-minus and does not establish priority.

For the correction in section 2.1, the checked artifact was arXiv:2210.09115v1 and the bound excerpt has SHA-256 `b7c4aaf6c75e5a1790fc17f311242a8c56d6d23fd153657baed7dd93421c022f`. The source audit records CORRECTION_NOT_NOVELTY_NOT_EXACT_DUPLICATE and version-of-record-line-checked = false. This source distinction is part of the claim, not an editorial aside.

B.3 Canonical artifact bindings

Table 2 is mechanically traceable to the following protected files. Paths are relative to the Paper-44 authority package.

Table 4: Protected source and result bindings used by the manuscript.

Relative path	SHA-256
`preauthority/SHA256SUMS.txt`	`1952daeee561e4b0e1d11795a9638803a288a1eecddab0702ebcfec95816a7fd`
`preauthority/SOURCE_LOCK.md`	`a49bbc392e21a25e7f36ab8c0c5426bbec510aa30bc6d2d6943b0e81c5347984`
`preauthority/PROOF_PACKAGE.md`	`9367109c025c885c11f2e49b9bdef0353b867efd618eb4698f171be8161757e0`
`preauthority/LITERATURE_NOVELTY_AUDIT.md`	`1de200d9757fab8107bc5d11791c7a903034e97307d27f060a1b6b07b04130f0`
`preauthority/OBJECT_MARKER_OPERATOR_CONTRACT.md`	`059ecbc4edcd097cb9eb83a0452591735fc25ca6d6d8da5a3ce10f4cff15330f`

Relative path	SHA-256
`outputs/reports/EXPERIMENT_REPORT.md`	`a41d0e856d6f0473714f7c3e905e44a0b6832da0db9937ac07b16e8ce6d5a1e9`
`outputs/results/exact_comparison.json`	`c988d083cbdd05a84f698497bfb6b8c13f9540fe3801497da9c7ce05d09f8e10`
`outputs/RESULT_LEDGER.json`	`2397a01fb66e42980b9a71a6861350afe52ceb258d3ae119dcbeef58b4d3086e`
`outputs/evaluations/route_a/SD-C46/2026-08-18.yaml`	`f0185ab27834d8ebfe158c57cf8a25a46e745493d5a331ea459e582ebe4835e3`
`outputs/audits/proof_audit.json`	`afd17de5efd811a90645efaf5aadf96cec0881a6b73d25966c1b081204708cde`
`outputs/audits/source_audit.json`	`9fffd522fe108d3c6cf9580a835757192020abbe967ef61354d1a1a8314ea0a3`
`outputs/audits/type_audit.json`	`966dee537ef56ae904ea09f9c8383d2f4092bc043224fff2b0fba436cf200e08`
`outputs/audits/independence_audit.json`	`d9eb6ab3e767654e84d723bbe09ad21ddceccd8079ff2c042ab8584e53822a51`
`outputs/audits/route_primary.json`	`6d6d92202f299fa38328a156c9e0874d5fe649312b55403e25d845297c79eba7`
`outputs/audits/route_independent.json`	`95136972f0c9d0b554f3d7920bc470771bcb6c8cea4fa16a24880d8aec3f7b8e`
`outputs/audits/integrity_audit.json`	`66ae3af081047fa9f896346a1d0c4698b933f85de0812188b3df79db78817b86`
`outputs/tests/mutation_results.json`	`21f9238294305d6adcff2c0afa66576c878f8a3d5b964b540a2372872261f9d6`
`outputs/audits/external_auditor_mutations.json`	`32b37ae129c606fba2f12826379295232188809f0e75478b3c3149f37385a65f`

B.4 Protected-input and writer chronology

Before any writer-candidate file was created, all 62 regular files in the post-output authority package were enumerated with relative path, kind, mode, UID, GID, size, modification time, inode, link count, and SHA-256. There were no symbolic links. The resulting protected snapshot has SHA-256 `a364048f5be1f9b88dedae5cde2f92d69295e4f403d92da310bdda301479539a`. The repository baseline was commit `6e5658649d2eab0fce077cbcdcc00070dd54095f`; the Paper-44 authority directory was an untracked tree at that moment. Writing, review, figure generation, and compilation occur only in a temporary candidate directory.

Candidate selection was retrospective. An earlier finite-prime-square periodic census and an all-k perfect-power operator candidate were stopped after source ownership was subtracted. The present position was assigned only after the exact remainder and golden controls were already known. Numerical adjacency among candidate or paper labels carries no scientific meaning.

B.5 Reproducibility and disclosure

The canonical computations are CPU-only exact integer, rational, algebraic, and outward-rounded interval checks; there is no model training, target fitting, stochastic seed, or GPU dependence. A reproduction must use a disposable copy of the sealed package and must not run in the protected authority tree. The State-A integration command, namespace, and hostile environment checks are documented in the protected package README. The paper reports stored outputs rather than rerunning or modifying them.

Bibliographic metadata were checked against the DOI records and the frozen source audit; the six entries in `references.bib` are the six entries cited by the manuscript. Automated language-model reviews were used to stress-test the outline and presentation. They supplied no theorem premise, proof step, experimental value, source priority, or authorization claim.

References

[1] Jung-Chao Ban, Wen-Guei Hu, and Song-Sun Lin. Pattern generation problems arising in multiplicative integer systems. *Ergodic Theory and Dynamical Systems*, 39(5):1234–1260, 2019. doi: 10.1017/etds.2017.74.

[2] Jung-Chao Ban, Wen-Guei Hu, and Guan-Yu Lai. Boundary complexity and surface entropy of 2-multiplicative integer systems on $\ltimes^d$. *Journal of Mathematical Physics*, 64(6):062704, 2023. doi: 10.1063/5.0118652. Author manuscript v1 submitted 17 October 2022.

[3] Jung-Chao Ban, Wen-Guei Hu, Guan-Yu Lai, and Lingmin Liao. Hausdorff dimensions of affine multiplicative shifts. *Advances in Mathematics*, 471:110266, 2025. doi: 10.1016/j.aim.2025.110266.

[4] Ai-Hua Fan, Lingmin Liao, and Ji-Hua Ma. Level sets of multiple ergodic averages. *Monatshefte für Mathematik*, 168(1):17–26, 2012. doi: 10.1007/s00605-011-0358-5.

[5] Richard Kenyon, Yuval Peres, and Boris Solomyak. Hausdorff dimension for fractals invariant under multiplicative integers. *Ergodic Theory and Dynamical Systems*, 32(5):1567–1584, 2012. doi: 10.1017/S0143385711000538.

[6] Manfred G. Madritsch. The summatory function of q-additive functions on pseudo-polynomial sequences. *Journal de Théorie des Nombres de Bordeaux*, 24(1):153–171, 2012. doi: 10.5802/jtnb.791.